Pratik Kalshetti

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SUMMARY

Researcher and **Systems Engineer** with expertise in **3D Computer Vision** and **Computer Graphics**.

Experienced systems architect in designing computer vision algorithms for custom **hardware accelerator** (GPU/NPU/ASIC) in AR/VR, Robotics, and Autonomous Driving.

Core research on topics including *Neural Reconstruction*, *Gaussian Splatting*, *Inverse Rendering*, *Pose Estimation*, and *SLAM*, with first-author papers published at top conferences/journals, including SIGGRAPH and TPAMI, and recognized by multiple research awards and industry fellowships.

EDUCATION

PhD in Computer Science and Engineering Jul 2018 – May 2024
Indian Institute of Technology Bombay (IIT Bombay)

Thesis: Reconstructing Hand Shape and Appearance for Accurate Tracking from Monocular Video
(Presentation at SIGGRAPH Asia Doctoral Consortium: https://youtu.be/nhGa4PcwmSE)

Advisor: Prof. Parag Chaudhuri

Honors:

- Two-time top-three finalist in ACM Student Research Competition at SIGGRAPH (2019, 2022)
- Research featured on ACM SIGGRAPH Blog
- TCS Research Scholar Fellow

Master of Technology (MTech) in Computer Science and Engineering Jul 2016 – May 2018
Indian Institute of Technology Bombay (IIT Bombay)

CPI: 9.37/10
Honors: Winner of Qualcomm Innovation Fellowship

Bachelor of Technology (BTech) in Computer Science and Engineering Jul 2012 – May 2016
Indian Institute of Technology Jodhpur (IIT Jodhpur)

CPI: 8.32/10
Honors: Best B.Tech. Project

EXPERIENCE

Qualcomm, Senior Lead Engineer Aug 2024 – Present

- Systems architect for designing 3D Reconstruction algorithms for custom hardware accelerator (ASIC).
- Owned and led the systems development of a new algorithm block for an upcoming VR chipset.
- Improved the algorithm, leading to substantial reduction in area ($\downarrow 98\% \mu m^2$) and cycle count ($\downarrow 37\%$).
- Collaborate with hardware, firmware, and software teams located across various geographies.
- Mentored cross-functional teams on computer graphics fundamentals and 3D reconstruction.

MPI-INF and **IIT Bombay**, Postdoctoral Research Fellow May 2024 – Jul 2024

- *Collaborator:* Dr. Vladislav Golyanik
- *Project:* High frame rate tracking of human hands from an event camera.
- *Idea:* 3D Gaussian splatting with Real-time Detection Transformers for fast hand reconstruction.

Adobe Research, Research Intern Jul 2021 – Nov 2021

- *Collaborators:* Alec Jacobson, Oliver Wang, Thibault Groueix, Vova Kim
- *Project:* Efficient video editing with high-level cues.
- *Idea:* Propagate frame-level edits via texture registration using proposed mesh optical flow.

PUBLICATIONS

- **Pratik Kalshetti** and Parag Chaudhuri. HandRT: Simultaneous Hand Shape and Appearance Reconstruction With Pose Tracking From Monocular RGB-D Video. In *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2025. [Paper] [Code]
- **Pratik Kalshetti** and Parag Chaudhuri. Intrinsic Hand Avatar: Illumination-aware Hand Appearance and Shape Reconstruction from Monocular RGB Video. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024. [Paper] [Code] [Video]
- **Pratik Kalshetti**. Reconstructing Hand Shape and Appearance for Accurate Tracking from Monocular Video. In *SIGGRAPH Asia Doctoral Consortium*, 2023. [Paper] [Poster] [Video]
- **Pratik Kalshetti** and Parag Chaudhuri. Local Scale Adaptation to Hand Shape Model for Accurate and Robust Hand Tracking. In *Computer Graphics Forum (ACM SIGGRAPH/Eurographics Symposium on Computer Animation)*, 2022. [Paper] [Code]
- **Pratik Kalshetti** and Parag Chaudhuri. Local scale adaptation for augmenting hand shape models. In *SIGGRAPH Posters*, 2022. [Paper] [Blog]
- **Pratik Kalshetti** and Parag Chaudhuri. Unsupervised incremental learning for hand shape and pose estimation. In *SIGGRAPH Posters*, 2019. [Paper]
- **Pratik Kalshetti**, Manas Bundele, Parag Rahangdale, Dinesh Jangra, Chiranjoy Chattopadhyay Antara: An Interactive 3D Volume Rendering and Visualization Framework In *arXiv*, 2018. [Paper]
- **Pratik Kalshetti**, Manas Bundele, Parag Rahangdale, Dinesh Jangra, Chiranjoy Chattopadhyay, Gaurav Harit, Abhay Elhence. An Interactive Medical Image Segmentation Framework using Iterative Refinement. In *Computers in Biology and Medicine*, 2017. [Paper] [Project]

PROGRAMMING

- *General Programming:* Python, C++.
- *Machine Learning:* PyTorch, TensorFlow, Jax.
- *Graphics and Vision:* OpenGL, PyTorch3D, ModernGL, Open3D, OpenCV.
- *Linear Algebra:* Eigen, NumPy
- *Parallel Programming*: CUDA, OpenMP.
- *Development*: Linux, Git, CMake.

SERVICE

- Reviewer, IEEE Transactions on Image Processing (2025)
- Student volunteer at SIGGRAPH 2021, virtual
- Student volunteer at SIGGRAPH 2018, Vancouver
- Head, Public Relations and Hospitality, IGNUS 2015, IIT Jodhpur
- Coordinator, Electronics Club, IIT Jodhpur (2013)
- Student Guide, Counselling Service, IIT Jodhpur (2013)